



main.py



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Output

```
1 def bubble_sort_plain(arr):
13
14
15 def bubble_sort_optimized(arr):
16     arr = arr.copy()
17     comparisons = 0
18     n = len(arr)
19
20     for i in range(n):
21         swapped = False
22
23         for j in range(n - i - 1):
24             comparisons += 1
25
26             if arr[j] > arr[j + 1]:
27                 arr[j], arr[j + 1] = arr[j + 1], arr[j]
28                 swapped = True
29
30         if not swapped:
31             break
32
33     return arr, comparisons
```

All test cases passed!

=== Code Execution Successful ===